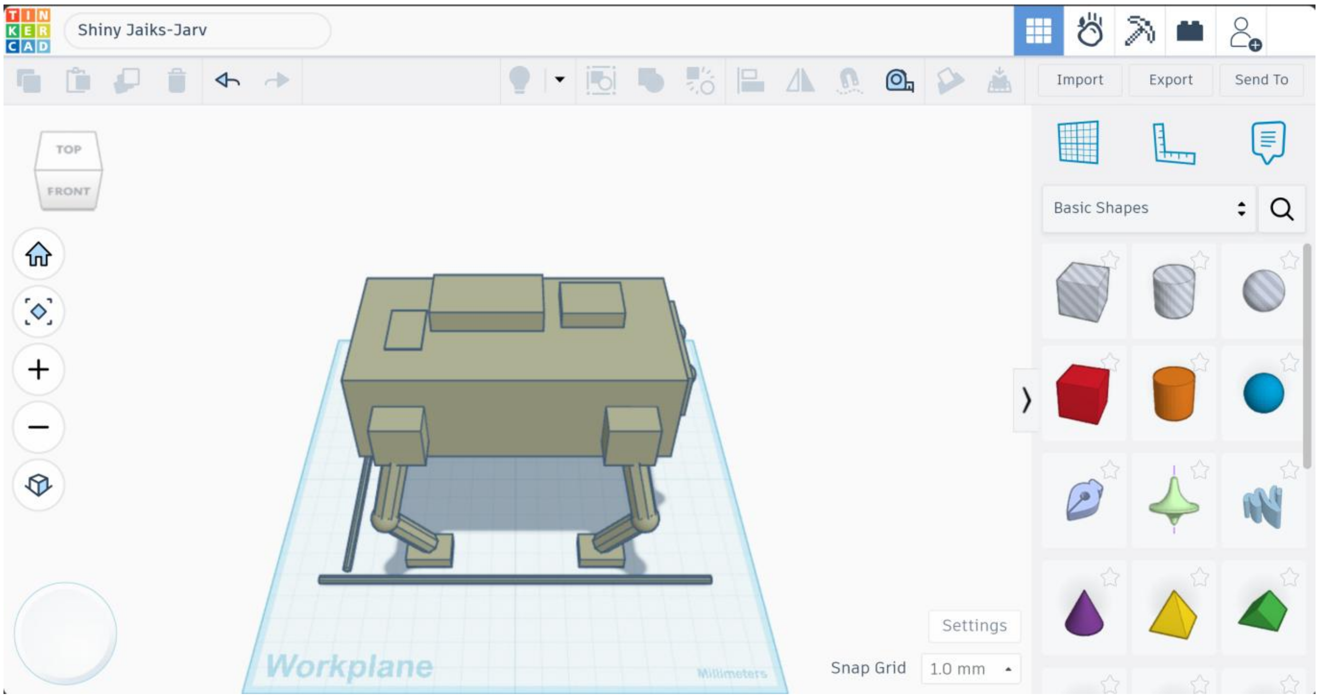
Initial Mechanical Design of a Simple Robotic Dog

Preliminary Design Report — Quadruped Robot

The goal of this design is not to build an advanced robot, but to understand the mechanical fundamentals that allow the robot to stand and walk with stability. Below is the proposed initial design, with calculations and engineering justifications for each element.



1. Body Shape and Chassis

A rectangular chassis approximately 20 x 10 cm is proposed, made of lightweight aluminum or 3-4 mm thick acrylic, balancing rigidity and low weight.

- Length: 20 cm — Width: 10 cm — Thickness: 3-4 mm.
- Suggested material: acrylic or lightweight aluminum, to minimize total load on the motors.
- Target total robot weight (excluding legs): approximately 1.5 kg.
- The four legs are distributed symmetrically (2 front + 2 rear) around the body center to ensure balanced loading.
- Mounting holes are added for the motors, battery, and control board, positioned as low as possible to keep the center of gravity low.

2. Leg Design

Each leg consists of two main segments (Femur - Tibia) in an inverted-Z configuration, the most common layout for simple quadruped robots because it provides good forward/backward range of motion while remaining easy to manufacture.

- Femur length: 6 cm.
- Tibia length: 6 cm.
- Total leg length: approximately 12 cm.
- The foot end is rounded or fitted with a small rubber foot pad to increase ground friction and absorb shocks.

3. Number of Joints and Degrees of Freedom (DOF)

Since a simple initial design is required, 2 degrees of freedom (2-DOF) per leg is proposed as the minimum for basic walking motion, with the option to upgrade to 3-DOF later for improved maneuverability.

Total Joints	DOF per Leg	Joint Type
8 joints	2 (hip/shoulder + knee)	Revolute
12 joints (optional upgrade)	3 (hip yaw + hip pitch + knee)	Revolute

- Baseline design: 8 joints total (one hip joint + one knee joint per leg, x4 legs).
- This is sufficient to generate a basic gait (Walk or Trot) without adding unnecessary control complexity.

4. Motor Selection

Servo motors are recommended over plain DC motors, since they provide precise angular position control, which is essential for coordinated walking.

- Suggested type: MG996R-class servo or equivalent (stall torque up to ~1.5 N·m at 6V).
- Number of motors required: 8 (one per joint in the baseline design).
- Operating voltage: 5-6V, with a separate power supply for the motors apart from the control board to avoid voltage drops.
- Choose motors with torque rated above the calculated requirement, with a safety factor of at least 1.5 (see calculation below).

5. Preliminary Torque Calculation for One Joint

To calculate the required torque for the hip joint as an example, the following assumptions are used:

1. Total robot mass: m = 3 kg.
2. Gravitational acceleration: g = 9.81 m/s^2.

3. Total weight: $W = m \times g = 3 \times 9.81 \approx 29.4$ N.
4. Worst case (trot gait): half the robot's weight rests on only two diagonal legs, so each leg carries: $F = W / 2 \approx 14.7$ N.
5. Torque arm length (distance from the joint to the foot's ground contact point): $L \approx 0.12$ m.
6. Required torque: $\tau = F \times L = 14.7 \times 0.12 \approx 1.76$ N·m.
7. Adding a safety factor of 1.5 to cover acceleration and vibration during motion: $\tau_{design} \approx 2.6$ N·m ($\approx$ 27 kg·cm).

Result: a servo motor with a torque rating of at least 27 kg·cm should be selected for the hip joint, while the knee joint can use a slightly lower torque rating since its load arm is typically smaller while standing.

6. Stability and Center of Gravity

- The center of gravity is placed at the exact middle of the body, at as low a height as possible (e.g., by mounting the battery low in the chassis).
- The center of gravity's ground projection must always remain inside the support polygon formed by the three or four ground-contact points of the legs.
- When one leg is lifted during walking, the projected center of gravity must remain inside the triangle formed by the three remaining legs (static stability).
- Weight distribution: front-to-back weight distribution should be kept close to even (45%-55%) to avoid the robot tipping forward or backward.

7. Proposed Walking Gait

For a simple design, it is recommended to start with a static walk (creep gait) before progressing to a trot gait:

- Creep gait: only one leg is lifted at a time while the other three remain on the ground, ensuring full stability, though the motion is slow.
- Suggested leg sequence: rear-right -> front-right -> rear-left -> front-left.
- Once this gait is verified to work reliably, the robot can transition to a trot gait (moving two diagonal legs together) for greater speed.

8. Expected Mechanical Issues

- Foot slipping on smooth surfaces due to insufficient friction — addressed by adding rubber foot pads.
- Joint slack or vibration caused by mechanical backlash in low-cost servo gears.
- Insufficient motor torque when the ground surface changes or the load increases suddenly.
- Motor overheating under sustained load over long periods, especially at the hip joint, which carries the highest torque.
- Loss of synchronization between the four legs' movements, which can cause instability if timing is not controlled precisely in software.

- Chassis sagging if the material used is too thin or insufficiently reinforced at the motor mounting points.

Summary: this initial design provides a realistic mechanical foundation that can be built with simple resources, and can later be extended by adding a third degree of freedom per leg or upgrading the motor type based on practical test results.